BIOMETRIC FATIGUE DOJI & MULTI-SPORT INTELLIGENCE

Technical Architecture, OHLC Japanese Candlestick Mechanics, & Multi-Sport Run of Show

Entity: National Ice Ball League Sport, Inc. (dba: BIM-APES Software / NIBLS, Inc.) | **Sole IP Owner:** Dashawn Ramel Bledsoe
Alternate Names: National Intelligent Ball League Sports | National Intelligent Blockchain League Sports
Operating Stack: Biometric Index Market (BIM-APES) / PhysioBioFin Platform | **Date:** October 2026

1. Executive Overview & Japanese Candlestick (OHLC) Engine

Traditional capital markets evaluate sports organizations and athlete name, image, and likeness (NIL) through lagging, subjective narrative sentiment. The **BIM-APES (Biometric Index Market)** and **PhysioBioFin** platform transforms live, non-invasive 1000Hz physiological telemetry (ECG/HRV, IMU kinematics, VO2, force velocity) directly into **Biometric Japanese Candlesticks (OHLC)** across discrete time windows (t).

Important Architectural Boundary: Per strict system design, this framework operates **exclusively on classic Japanese Candlestick geometry** (Open, High, Low, Close). All 3D volumetric spatial capsules, mesh overlays, and bounding blurs are completely omitted to ensure zero rendering latency and immediate sub-500ms execution across legacy broadcast feeds and venue displays.

Candlestick Parameter	Biometric Measurement Input	Physical & Financial Interpretation
Open (O_t)	Pre-Effort Readiness Score (PRI / CSI)	Baseline physical readiness and autonomic nervous system state at window start.
High (H_t)	Peak Force / Impact Load (IMU / GRF)	Maximum kinetic torque, G-force impact vector (>105 mph exit / >18 mph sprint).
Low (L_t)	Autonomic Strain / HRV Collapse	Maximum cardiovascular suppression, autonomic stress drop, and recovery drag.
Close (C_t)	Post-Effort Recovery Status (HRR)	Heart-rate recovery efficiency and physiological equilibrium at window close.

Fatigue Doji Trigger Mechanics

Mathematical Validation Formula:
$|\text{Close} - \text{Open}| \leq \delta \quad \text{AND} \quad (\text{High} - \text{Low}) \geq \gamma$

A **Fatigue Doji** occurs when the candlestick body compresses to a hairline ($|\text{Close} - \text{Open}| \leq \delta$) following a massive exertion wick ($\text{High} - \text{Low} \geq \gamma$). This signals a critical physiological impasse: maximum kinetic effort producing minimal mechanical output. Upon validation, the **EventRuleEngine** fires POST /api/sponsor/activation/track in **under 500ms**.

2. Valuation Engine: Model NIL vs. Market NIL & Arbitrage

The PhysioBioFin trading stack evaluates two distinct financial products simultaneously to locate arbitrage spreads:

• **Model NIL (Intrinsic Physical Capability):** Calculated at 1000Hz using Fama-MacBeth 2-pass regressions across 5 physiological risk factors (PMF-E, CSE-E, FBR-E, SEF-E, WD-CS-E). Fully isolated from market sentiment and macro interest rates.

• **Market NIL (Marketplace Order-Book Price):** Floating price determined on secondary order books by public crowd sentiment, media narrative, and order flows.

Market Regime	Dislocation Spread (Δ)	CPM Pricing & Surcharges	Quant Desk Action
Bull Regime	Model NIL > Market NIL	USD $1,200 – $2,000 (+40% Bull Surge)	Long Pair Trade / Call Option Purchase
Sideways Regime	Model NIL = Market NIL	USD $600 – $1,000 (Base Rate)	High-Frequency Scalping (<10ms Latency)
Bear Regime	Model NIL < Market NIL	USD $300 – $550 (+25% Risk Surcharge)	Put Option / Pay-Fixed NIL Swap

3. Programmatic 60/20/10/10 Clearing & Sponsor Margin Accounts

All gross in-game fan tipping, boost credit purchases, and QR flash-sale redemptions clear programmatically on-chain (Solana sub-second rails / Ethereum Reg D settlement) under the **60/20/10/10 Split Rule**:

60.0% → Athlete NIL Wallet	Direct creator pool disbursed instantly to athlete wallet.
20.0% → Team / League Pool	Liquidity allocation for organizational operations and venue logistics.
10.0% → Sponsor Royalty Bucket	Swept into Tier-1 Custodial Margin Accounts (Morgan Stanley GSE 3.0, J.P. Morgan, Capital One) delivering a **3.1x to 3.5x self-liquidating ROI cushion**.
10.0% → Platform Reserve	BIM-APES governance fee, protocol maintenance, and risk buffer.

4. Multi-Sport Realistic Scenario Run of Shows

To demonstrate real-time execution across diverse athletic environments, the following section details step-by-step 1000Hz signal ingestion, Doji pattern validation, sub-500ms API triggers, and 60/20/10/10 clearing across four professional sports disciplines:

A. Racing (Formula 1 / NASCAR / Motorsports)

Telemetry Ingestion (1000Hz): Sensors measure driver steering wheel peak torque (Nm), neck lateral G-force strain (up to 5.5G in high-speed corners), core thermal stress (°C), and brake pedal pressure deceleration.
Fatigue Doji Identification: On Lap 48 of 56, the driver experiences severe thermal accumulation. High Wick registers peak 5.2G cornering load, but Open and Close readiness scores converge into a compressed Doji body ($|Close - Open| \leq \delta$), indicating steering micro-drift and physical exhaustion.
Sub-500ms Webhook Execution: EventRuleEngine fires POST /api/sponsor/activation/track in 460ms. Telemetry overlays load on the global broadcast lower-third (*'Gatorade Hydration Strain Index: 94%'*) alongside pit-strategy sponsor tickers.
Fan Micro-Rail Clearing: Spectators scan TV QR codes to tip $2.00 into the driver's pit-crew boost pool. Total tip volume of $25,000 clears via 60/20/10/10: $15,000 to Driver, $5,000 to Racing Team, **$2,500 swept into Sponsor Margin Account**, and $2,500 to Platform Reserve.

B. Boxing (Championship Combat Sports)

Telemetry Ingestion (1000Hz): Smart glove sensors measure punch acceleration and G-force impact (g), mouthguard sensors record head rotational acceleration (rad/s²), and wearable telemetry tracks cardiac stress.
Fatigue Doji Identification: In Round 9, Fighter A lands a heavy combination (High Wick: 42g impact load), but autonomic recovery collapses (Low Wick). The candlestick body shrinks to a hairline line as heart-rate recovery stagnates at 178 BPM, confirming a Fatigue Doji reversal state.
Sub-500ms Webhook Execution: Webhook fires in 420ms. Pay-per-view broadcast displays a 2D side banner: *'Monster Energy Impact Alert & Stamina Threshold'* while mobile apps push $1.00 energy drink coupons.
Fan Micro-Rail Clearing: $40,000 in micro-tips clears programmatically: $24,000 to Fighter Wallet, $8,000 to Promoter Liquidity Pool, **$4,000 to Sponsor Royalty Account**, and $4,000 to Platform Reserve.

C. American Football (NFL / Playoff Championship)

Telemetry Ingestion (1000Hz): Shoulder pad IMUs record linear collision G-force, cleat sensor array measures acceleration bursts (m/s²), and telemetry tracks heart-rate variability during 4th-quarter two-minute drills.
Fatigue Doji Identification: Star Running Back carries 6 consecutive plays. High Wick records 18.5 mph sprint speed and a 12g tackle impact, but the candlestick body compresses to 0.4 points as HRV drops into the red zone, confirming autonomic overload.
Sub-500ms Webhook Execution: Webhook executes in 480ms. Stadium Jumbotron and TV lower-thirds flash: *'McDonald's Fuel the Drive Threshold'* alongside pulsing QR codes for $1.00 fries and Victory Meal app downloads.
Fan Micro-Rail Clearing: 12,500 fans scan QR codes generating $50,000 in micro-tips: $30,000 to Player Wallet, $10,000 to Franchise Pool, **$5,000 swept into McDonald's Custodial Margin Account**, and $5,000 to Platform Reserve.

D. Basketball (NBA / Game 7 Finals)

Telemetry Ingestion (1000Hz): Insole sensors measure vertical jump force (N) and landing impact, waist IMUs record lateral deceleration, and cardiac telemetry tracks real-time strain during double-overtime.
Fatigue Doji Identification: Point Guard plays 44 continuous minutes. High Wick registers a 38-inch vertical jump on a drive, but Open/Close readiness scores converge at a low plateau (88.4 -> 88.6), producing a classic Doji pattern signaling imminent jump-shot decay.
Sub-500ms Webhook Execution: Fired in 410ms. TV broadcast lower-third renders: *'Nike Speed Index Burst: 19.2 mph'* while pushing geofenced mobile app notifications.
Fan Micro-Rail Clearing: $30,000 in live fan boost credits clears: $18,000 to Player Wallet, $6,000 to Team Pool, **$3,000 swept into Nike Custodial Margin Account**, and $3,000 to Platform Reserve.

5. Quantitative Hedge Desk Execution (24-Method Highlights)

Upon confirming a Fatigue Doji pattern, institutional trading desks execute automated risk management protocols across the **PhysioBioFin 24-Method Execution Stack**:

Method 1: Biometric Adaptive Trailing Stop	Dynamically tightens position trailing stops as fatigue accumulates, locking in gains prior to public market sentiment decay.
Method 2: Static Pattern-Only Exit	Executes fixed exit rules based on Japanese Candlestick geometry independent of biometric inputs.
Method 4: Volatility Hard Threshold Stop	Triggers immediate position liquidation when biometric stress or trading volatility exceeds preset standard deviation bands.
Method 21: Collapse Prediction Model	Medical risk model predicting acute physical collapse or injury risk, executing defensive put options and medical alerts.

6. Formal Proof of Claim & Zero-Dilution IP Declaration

> **IP Legal Protection & Governance**
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> **Formal Declaration of Sole Ownership:**
> All software architectures, 1000Hz telemetry normalization algorithms, Japanese Biometric Candlestick (OHLC) Doji trigger conditions, 5 black-box rule engines (EventRuleEngine, BiometricRuleEngine, NILRuleEngine, OverlayRuleEngine, SponsorRuleEngine), and 60/20/10/10 clearing engines are **100% sole personal intellectual property of Dashawn Ramel Bledsoe** under **National Ice Ball League Sport, Inc. (dba: BIM-APES Software / NIBLS, Inc.)**.
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